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CS 490: Intensive Python

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Final Capstone Report

1. Introduction and Motivation

This project investigates which vehicle characteristics influence fuel efficiency in modern non-electric vehicles, and how fuel efficiency has changed over time. This question matters because fuel efficiency affects everyday transportation costs, environmental impact, and automotive design decisions. Consumers often seek more fuel-efficient vehicles to save on gasoline costs or to reduce their carbon footprint. Because of this, it is important to know which automobile characteristics have the largest effect on fuel efficiency when purchasing a vehicle.

To understand these relationships, this project uses the *Vehicle MPG (1984-2023)* dataset downloaded from Kaggle. This dataset contains detailed information on nearly 46,000 vehicles, 26 columns covering features such as manufacturer, miles per gallon (city, highway, and combined), and fuel type, on vehicles built between 1984 and 2023. For this analysis, the dataset will be filtered to focus on modern vehicles from 2000 onward and will exclude electric vehicles to ensure consistency in fuel-efficiency measurements. By examining this data, this project aims to identify the most important factors influencing fuel efficiency and evaluate how these factors have contributed to improvements over time.

2. Data Description and Cleaning

The dataset used in this project contains information on approximately 46,000 vehicles and includes a variety of features related to vehicle specifications and fuel efficiency. Key

variables used in this analysis include both numerical and categorical data and are listed in the table below.

Variable Category	Specific Fields
Numerical	Model Year, Miles per Gallon (City, Highway, and Combined), Engine Displacement, and Engine Cylinders
Categorical	Fuel type, Transmission, Drive (drivetrain configuration), and Vehicle Class

Before cleaning, I performed initial data quality checks by inspecting data structure, reviewing data types, and counting missing values. I also checked for duplicate entries to ensure there were no repeated records. Initial inspection showed missing values in several key variables, particularly engine characteristics and fuel efficiency metrics. Additionally, electric vehicles introduced inconsistencies because they do not use the traditional miles-per-gallon measurement. These complications would need to be addressed before proceeding with deeper analysis.

To accommodate these findings, I filtered the dataset to include only vehicles from the year 2000 and later, per my focus on modern vehicles. I also removed electric vehicles by identifying rows with electric fuel type or electric vehicle range values. Removing electric vehicles simplified this analysis and helped avoid complications further on. Next, I retained only the variables relevant to fuel efficiency and modeling by removing unnecessary columns, such as identifiers and descriptive fields, that do not contribute to predicting fuel efficiency. I then removed rows with missing values in key variables, particularly the Combined MPG, to ensure the dataset was complete for modeling.

After these cleaning steps, the dataset was reduced to roughly 27,000 rows, leaving a more focused subset of variables and observations that would influence my initial question. This cleaned dataset provides a consistent and reliable foundation for exploratory analysis and regression modeling.

3. Exploratory Analysis

Before exploring relationships between variables, I first examined the overall distribution of the dataset. Figure 1 presents summary statistics for key variables in the dataset after cleaning.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
model_year	26942.0	NaN	NaN	NaN	2011.98237	6.74634	2000.0	2006.0	2012.0	2018.0	2023.0
Make	26942	75	Chevrolet	2056	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Model	26942	3406	Jetta	144	NaN	NaN	NaN	NaN	NaN	NaN	NaN
fuel_type	26942	5	Regular Gasoline	14580	NaN	NaN	NaN	NaN	NaN	NaN	NaN
city_mpg	26942.0	NaN	NaN	NaN	18.905761	5.582707	7.0	15.0	18.0	21.0	58.0
highway_mpg	26942.0	NaN	NaN	NaN	25.54094	5.96907	11.0	21.0	25.0	29.0	61.0
combined_mpg	26942.0	NaN	NaN	NaN	21.34307	5.650552	8.0	18.0	21.0	24.0	59.0
cylinders	26942.0	NaN	NaN	NaN	5.802836	1.829229	2.0	4.0	6.0	6.0	16.0
displacement	26942.0	NaN	NaN	NaN	3.299031	1.31491	0.6	2.2	3.0	4.0	8.4
transmission	26942	34	Automatic 4-spd	3918	NaN	NaN	NaN	NaN	NaN	NaN	NaN
drive	26942	7	Front-Wheel Drive	8510	NaN	NaN	NaN	NaN	NaN	NaN	NaN
vehicle_class	26942	26	Compact Cars	3658	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Figure 1: Statistical Summary of Key Variables

Then, I wanted to better understand the relationship between vehicle characteristics and fuel efficiency, so exploratory data analysis (EDA) was conducted using several visualizations. The distribution of combined miles per gallon provides insight into the overall efficiency of modern vehicles. Combined MPG is used because it provides the most accurate representation of overall fuel efficiency compared to city or highway MPG alone. As shown in Figure 2, most vehicles fall within the range of 18-25 miles per gallon, with the mean being 21.34 MPG and the median at 21.0 MPG.

To further examine the relationship between engine characteristics and fuel efficiency, engine displacement was compared to the combined MPG. As shown in Figure 3, there is a strong negative relationship between

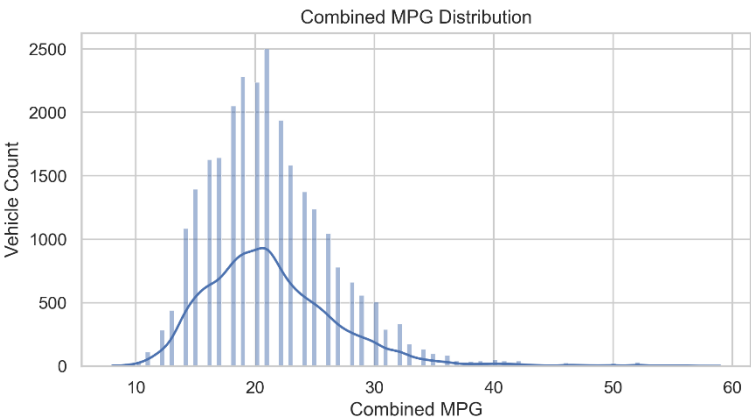


Figure 2: Distribution of Combined MPG in non-electric vehicles (2000 and later)

engine displacement and fuel efficiency. Vehicles with larger engines tend to have lower MPG, while vehicles with smaller engines tend to achieve higher fuel efficiency. Although there is some variability, the overall trend strongly supports that engine size is a major factor influencing fuel consumption.

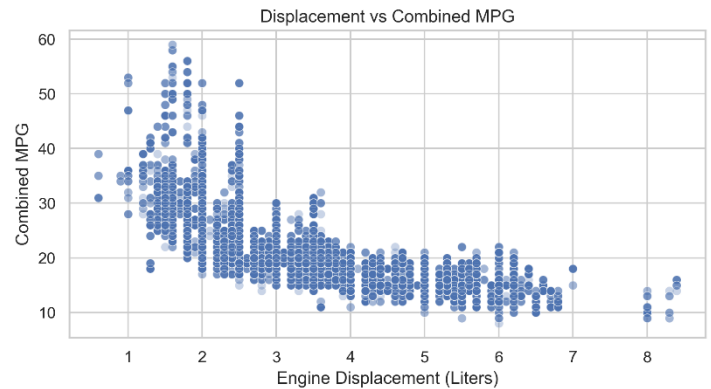


Figure 3: Engine Displacement vs Combined MPG in non-electric vehicles (2000 and later)

Upon learning that engine displacement is a large factor in fuel efficiency, I explored the relationship between the number of engine cylinders and MPG. This relationship resembled that of engine displacement and combined MPG, where fewer engine cylinders typically lead to better fuel efficiency. Three cylinders had the best overall fuel efficiency, and engines with more cylinders tended to have worse fuel efficiency. Despite these averages, four and six-cylinder engines had many outliers well above the average. This was expected, however, since those are some of the most common cylinder counts, resulting in a wider range of observed values.

Finally, a correlation matrix was used to summarize relationships among key numerical variables. Figure 5, the correlation matrix, shows a positive relationship between model year and miles per gallon. In contrast, the number of cylinders and engine displacement showed strong negative relationships with city, highway, and combined miles per gallon, reinforcing observed trends in previous figures. City, highway, and combined

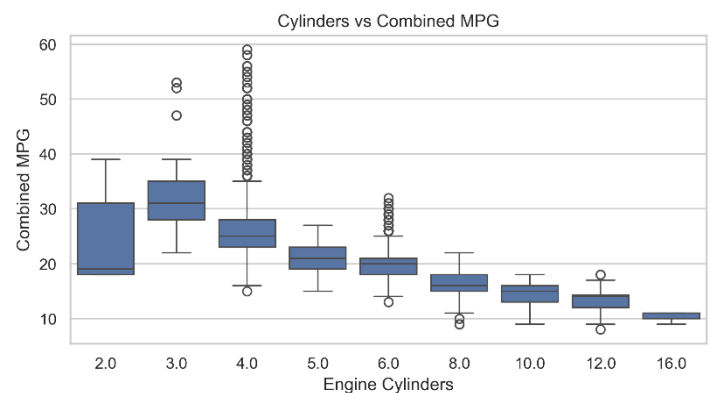


Figure 4: Cylinder count vs MPG in non-electric vehicles (2000 and later)

MPG show a strong positive relationship because combined MPG is calculated directly from city and highway miles per gallon.

4. Methods and Results

To model fuel efficiency, regression techniques were used since the target variable, combined MPG, is a continuous numerical value affected by multiple input features. Two models were implemented: a Linear Regression model as a simple baseline and a Random Forest model to capture potential non-linear relationships and interactions between variables.

To begin, a preprocessing pipeline was applied to prepare the data. Numerical features were handled using median imputation to address any remaining missing values, while categorical features were imputed using the most frequent value and then one-hot encoded. The dataset was then split into training and testing sets. Eighty percent of the data was used for training, while the remaining twenty percent was used to evaluate performance. Model performance was evaluated using three regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). These metrics provide insight into both the accuracy of predictions and the model's ability to explain variation in fuel efficiency. The results of the models are as follows:

	MAE	RMSE	R^2
Linear Regression	1.723	2.628	0.798
Random Forest	0.833	1.485	0.935

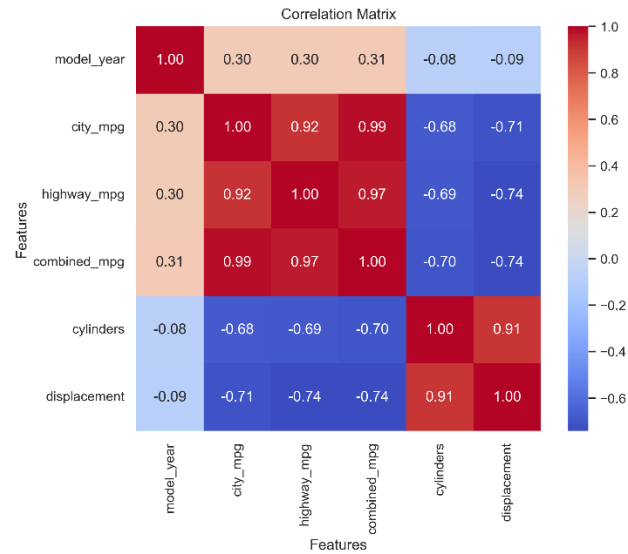


Figure 5: Correlation Matrix of Key Numerical Features

The Random Forest model outperformed the Linear Regression model across all three metrics. In particular, the lower RMSE and MAE values indicate more accurate predictions, while the higher R^2 value shows that the model explains a greater proportion of variance in fuel efficiency. This suggests that non-linear relationships and interactions between features play an important role in predicting MPG.

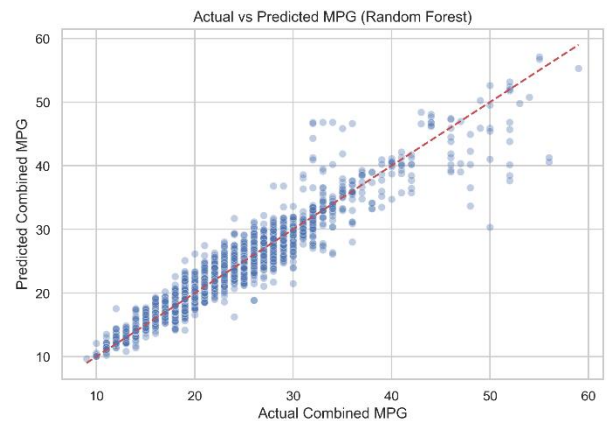


Figure 6: Random Forest Actual vs Predicted MPG Results

Figure 6 shows the relationship between actual and predicted MPG values for the Random Forest model. The points closely follow the diagonal reference line, indicating that the model can produce accurate predictions across a wide range of MPG values. Some dispersion is present at higher MPG values, but overall, the model demonstrates strong predictive performance.

Figure 7 presents the distribution of residuals, the differences between actual and predicted MPG. The residuals are centered around zero and appear approximately normally distributed, which suggests that the model does not exhibit significant bias, and prediction errors are relatively small.

Overall, the Random Forest model provides a strong fit for the data and significantly improves the accuracy compared to the baseline linear regression model. These results indicate that more complex

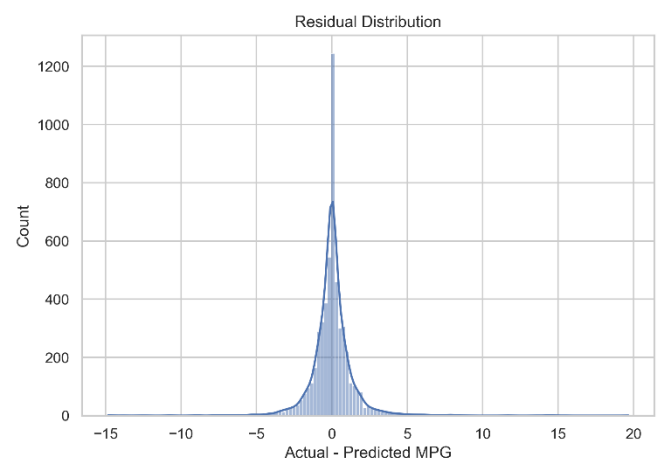


Figure 7: Residual Distribution of MPG in electric vehicles (2000 and later)

models are better suited for capturing relationships between vehicle characteristics and fuel efficiency.

5. Discussion and Conclusions

This project aimed to determine which vehicle characteristics most influenced fuel efficiency in modern non-electric vehicles and how fuel efficiency changed over time. Based on my analysis, engine displacement and the number of cylinders were found to be the strongest predictors of fuel efficiency, with both variables having clear negative relationships with combined MPG. Vehicles with larger engines and more cylinders generally achieve lower MPG. In contrast, model year showed a positive relationship with MPG, suggesting that fuel efficiency has improved over time, likely due to advances in automotive technology and efficiency standards.

Despite these findings, several limitations should be considered. This dataset is observational, so relationships identified cannot be interpreted as strictly causal. Additionally, the dataset does not include potentially important variables such as vehicle weight, aerodynamics, or driving behavior, which may also influence fuel efficiency. There is also a potential bias in the dataset, as it only reflects included vehicles, despite the large sample size, and may not fully represent all vehicles on the road.

Future work could improve this analysis by incorporating additional features and refining the modeling process. Techniques such as cross-validation could further improve model performance. Additionally, analyzing subsets of the data, such as by vehicle class, may provide detailed insights into how fuel efficiency varies across different types of vehicles.